STUDENT ATTENDANCE TRACKER

1. Student Details

Student Name	A.AKASH
Register Number	2513206180252100l
Department	B.Sc. Computer Science
Collegel	THIRU.A.GOVINDASAMY GOVERNMENT ARTS COLLEGE TINDIVANAM -604 307
ID	9994154546
Date /submission	24/09/2026

2. Project Details

Project Title	Student Attendance Tracker
Project Type	Responsive Web Application
Purpose	Digital attendance recording and monitoring
Target Users	Faculty / Teachers
Technologies	HTML, CSS, JavaScript, Local Storage

3. Problem Statement

Develop a responsive web application that enables faculty to record and monitor student attendance digitally. The system should make attendance marking, percentage calculation, student searching, and attendance history easier and more organized.

4.SOURCE CODE

```
HTML<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Attendance Tracker</title>

    <!-- Separate CSS File -->
    <link rel="stylesheet" href="style.css">
</head>

<body>

<!-- Header -->
<header>
    <h1>Student Attendance Tracker</h1>
    <p>College Attendance Management System</p>
</header>

<!-- Navigation -->
<nav>
    <a href="#home">Home</a>
    <a href="#add-student">Add Student</a>
    <a href="#attendance">Mark Attendance</a>
    <a href="#search">Search Student</a>
    <a href="#history">Attendance History</a>
    <a href="#reports">Reports</a>
</nav>
```

```
<!-- Home / Dashboard -->
<main>

    <section id="home">
        <h2>Dashboard</h2>

        <div class="dashboard">
            <div class="card">
                <h3>Total Students</h3>
                <p id="totalStudents">0</p>
            </div>

            <div class="card">
                <h3>Present</h3>
                <p id="presentStudents">0</p>
            </div>

            <div class="card">
                <h3>Absent</h3>
                <p id="absentStudents">0</p>
            </div>

            <div class="card">
```

```
            <h3>Average Attendance</h3>
            <p id="averageAttendance">0%</p>
        </div>
    </div>
</section>

<!-- Add Student -->
<section id="add-student">
    <h2>Add Student Details</h2>

    <form id="studentForm">

        <label for="studentName">Student Name</label>
        <input type="text" id="studentName" name="studentName"
            placeholder="Enter student name" required>

        <label for="registerNumber">Register Number</label>
        <input type="text" id="registerNumber" name="registerNumber"
            placeholder="Enter register number" required>

        <label for="department">Department</label>
        <input type="text" id="department" name="department"
            placeholder="Enter department" required>
```

```
        <label for="className">Class</label>
        <input type="text" id="className" name="className"
            placeholder="Enter class" required>

        <button type="submit">Add Student</button>

    </form>
</section>

<!-- Mark Attendance -->
<section id="attendance">
    <h2>Mark Attendance</h2>

    <div class="attendance-controls">
        <label for="attendanceDate">Date</label>
        <input type="date" id="attendanceDate">
        <button type="button" id="markPresent">
            Mark Present
        </button>

        <button type="button" id="markAbsent">
            Mark Absent
        </button>
```

```
</div>


<table>
  <thead>
    <tr>
      <th>S.No</th>
      <th>Student Name</th>
      <th>Register Number</th>
      <th>Department</th>
      <th>Status</th>
    </tr>
  </thead>

  <tbody id="attendanceTableBody">
    <!-- Attendance data will appear here -->
  </tbody>
</table>
</section>


<!-- Search Student -->
<section id="search">
  <h2>Search Student</h2>

  <form id="searchForm">
```

```
<label for="searchStudent">
   Student Name / Register Number
</label>

<input type="search"
      id="searchStudent"
      placeholder="Search student">
   <button type="submit">Search</button>

</form>

<div id="searchResult">
   <!-- Search result will appear here -->
</div>
</section>

<!-- Attendance History -->
<section id="history">
   <h2>Attendance History</h2>

   <table>
      <thead>
```

```
      <tr>
        <th>Date</th>
        <th>Student Name</th>
        <th>Register Number</th>
        <th>Status</th>
        <th>Attendance Percentage</th>
      </tr>
    </thead>
    <tbody id="historyTableBody">
      <!-- Attendance history will appear here -->
    </tbody>
  </table>
</section>

<!-- Reports -->
<section id="reports">
  <h2>Attendance Reports</h2>

  <div class="report-box">
    <h3>Student Attendance Report</h3>

    <p>
      View student attendance details,
```

```
            attendance percentage and attendance status.
        </p>


        <button type="button" id="generateReport">
            Generate Report
        </button>
    </div>
</section>


</main>


<!-- Footer -->
<footer>
    <p>
        © 2026 Student Attendance Tracker |
        College Project
    </p>
</footer>


    <!-- Separate JavaScript File -->
    <script src="script.js"></script>


</body>
</html>
```

```
CSS/* ================================

   Student Attendance Tracker

   CSS File

   ================================ */


* {

    margin: 0;

    padding: 0;

    box-sizing: border-box;

}


html {

    scroll-behavior: smooth;

}


body {

    font-family: Arial, sans-serif;

    background: #f4f7fb;

    color: #222;

    line-height: 1.6;

}


/* Header */

header {
```

```
    background: linear-gradient(135deg, #1e3c72, #2a5298);
    color: white;
    text-align: center;
    padding: 35px 20px;
}

header h1 {
    font-size: 32px;
    margin-bottom: 8px;
}

header p {
    font-size: 17px;
}

/* Navigation */
nav {
    background: #ffffff;
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 8px;
    padding: 15px;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
```

```
    position: sticky;
    top: 0;
    z-index: 100;
}

nav a {
    text-decoration: none;
    color: #1e3c72;
    font-weight: bold;
    padding: 10px 16px;
    border-radius: 6px;
    transition: 0.3s;
}

nav a:hover {
    background: #1e3c72;
    color: white;
}

/* Main */
main {
    width: 92%;
    max-width: 1200px;
    margin: 30px auto;
```

```
}


section {
    background: white;
    padding: 25px;
    margin-bottom: 30px;
    border-radius: 12px;
    box-shadow: 0 3px 12px rgba(0, 0, 0, 0.08);
}


section h2 {
    color: #1e3c72;
    margin-bottom: 20px;
    border-left: 5px solid #2a5298;
    padding-left: 12px;
}


/* Dashboard */
.dashboard {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 20px;
}
```

```
.card {
    background: linear-gradient(135deg, #2a5298, #1e3c72);
    color: white;
    padding: 25px;
    border-radius: 12px;
    text-align: center;
    transition: 0.3s;
}

.card:hover {
    transform: translateY(-5px);
}

.card h3 {
    font-size: 17px;
    margin-bottom: 10px;
}

.card p {
    font-size: 30px;
    font-weight: bold;
}

/* Forms */
```

```
form {
    display: grid;
    gap: 10px;
}

label {
    font-weight: bold;
    margin-top: 5px;
}

input,
select {
    width: 100%;
    padding: 12px;
    border: 1px solid #ccc;
    border-radius: 6px;
    font-size: 15px;
    outline: none;
}

input:focus,
select:focus {
    border-color: #2a5298;
}
```

```
/* Buttons */
button {
    background: #1e3c72;
    color: white;
    border: none;
    padding: 12px 20px;
    border-radius: 6px;
    font-size: 15px;
    font-weight: bold;
    cursor: pointer;
    margin-top: 8px;
    transition: 0.3s;
}

button:hover {
    background: #2a5298;
    transform: translateY(-1px);
}

/* Attendance Controls */
.attendance-controls {
    display: flex;
    align-items: center;
```

```
    gap: 12px;
    flex-wrap: wrap;
    margin-bottom: 20px;
}

.attendance-controls input {
    width: auto;
}

/* Tables */
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
    background: white;
}

th {
    background: #1e3c72;
    color: white;
    padding: 13px;
    text-align: center;
}
```

```
td {
    padding: 12px;
    text-align: center;
    border-bottom: 1px solid #ddd;
}

tr:hover {
    background: #f1f5fb;
}

/* Search Result */
#searchResult {
    margin-top: 20px;
    padding: 15px;
    min-height: 30px;
}

/* Report Box */
.report-box {
    background: #f4f7fb;
    padding: 25px;
    border-radius: 10px;
    text-align: center;
}
```

```
.report-box h3 {
    color: #1e3c72;
    margin-bottom: 10px;
}

.report-box p {
    margin-bottom: 10px;
}

/* Footer */
footer {
    background: #1e3c72;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 30px;
}

footer p {
    font-size: 14px;
}

/* Responsive Design */
```

```
@media (max-width: 900px) {

    .dashboard {
        grid-template-columns: repeat(2, 1fr);
    }

    nav {
        gap: 5px;
    }

    nav a {
        font-size: 14px;
        padding: 8px 12px;
    }
}

@media (max-width: 600px) {

header h1 {
    font-size: 24px;
}

header p {
    font-size: 14px;
```

```
}

main {
    width: 95%;
    margin: 20px auto;
}

section {
    padding: 18px;
}

.dashboard {
    grid-template-columns: 1fr;
}

nav {
    position: static;
    flex-direction: column;
    align-items: stretch;
}

nav a {
    text-align: center;
}
```

```
table {
    display: block;
    overflow-x: auto;
    white-space: nowrap;
}

.attendance-controls {
        flex-direction: column;
        align-items: stretch;
    }

    .attendance-controls input {
        width: 100%;
    }
}
```

JS

```
/* ========================================

   Student Attendance Tracker

   JavaScript File

   ======================================== */

let students = JSON.parse(localStorage.getItem("students")) || [];
```

```
let attendanceHistory =
    JSON.parse(localStorage.getItem("attendanceHistory")) || [];

let selectedAttendance = {};

/* ================================
   Save Data
   ================================ */

function saveData() {
    localStorage.setItem("students", JSON.stringify(students));
    localStorage.setItem(
        "attendanceHistory",
        JSON.stringify(attendanceHistory)
    );
}

/* ================================
   Add Student
   ================================ */

const studentForm = document.getElementById("studentForm");

if (studentForm) {
```

```
studentForm.addEventListener("submit", function (event) {

    event.preventDefault();

    const name = document.getElementById("studentName").value.trim();
    const registerNumber =
        document.getElementById("registerNumber").value.trim();
    const department =
        document.getElementById("department").value.trim();
const className =
    document.getElementById("className").value.trim();

if (!name || !registerNumber || !department || !className) {
    alert("Please enter all student details.");
    return;
}

const existingStudent = students.find(
    student => student.registerNumber === registerNumber
);

if (existingStudent) {
    alert("Register number already exists.");
    return;
```

```
}


const student = {
    id: Date.now(),
    name: name,
    registerNumber: registerNumber,
    department: department,
    className: className,
    totalClasses: 0,
    presentClasses: 0,
    absentClasses: 0
};

students.push(student);

saveData();
    studentForm.reset();

    alert("Student added successfully!");

    displayStudents();
    updateDashboard();
});
```

```
}

/* ==================================
   Display Students
   ================================== */

function displayStudents() {
    const tableBody =
        document.getElementById("attendanceTableBody");

    if (!tableBody) return;

    tableBody.innerHTML = "";

    students.forEach((student, index) => {
        const row = document.createElement("tr");

        row.innerHTML = `
            <td>${index + 1}</td>
            <td>${student.name}</td>
            <td>${student.registerNumber}</td>
```

```
<td>${student.department}</td>
<td>
  <button
    type="button"
    onclick="markAttendance(${student.id}, 'Present')">
    Present
  </button>
  <button
    type="button"
    onclick="markAttendance(${student.id}, 'Absent')">
    Absent
  </button>
</td>
`;

tableBody.appendChild(row);
  });
}

/* ================================
   Mark Attendance
   ================================ */
```

```
function markAttendance(studentId, status) {

    const dateInput =

        document.getElementById("attendanceDate");

    let date = dateInput ? dateInput.value : "";

    if (!date) {

        const today = new Date();

        date = today.toISOString().split("T")[0];

}

const student =

    students.find(item => item.id === studentId);

if (!student) {

    return;

}

const alreadyMarked = attendanceHistory.find(

    record =>

        record.studentId === studentId &&

        record.date === date
```

```
);

if (alreadyMarked) {

    alert("Attendance already marked for this date.");

    return;
}

student.totalClasses++;


if (status === "Present") {
    student.presentClasses++;
} else {
    student.absentClasses++;
}

attendanceHistory.push({
    studentId: student.id,
    name: student.name,
    registerNumber: student.registerNumber,
    date: date,
    status: status
});
```

```
    saveData();


    displayStudents();

    displayHistory();

    updateDashboard();


    alert(
        student.name +
        " marked as " +
        status +
        " on " +
        date
    );
}


/* ================================
   Attendance History
   ================================ */

function displayHistory() {


    const historyBody =
        document.getElementById("historyTableBody");
```

```
if (!historyBody) return;

historyBody.innerHTML = "";

attendanceHistory.forEach(record => {
  const student =
    students.find(item => item.id === record.studentId);

  let percentage = 0;

  if (student && student.totalClasses > 0) {
    percentage =
      (student.presentClasses /
        student.totalClasses) * 100;
  }

  const row = document.createElement("tr");

  row.innerHTML = `
    <td>${record.date}</td>
    <td>${record.name}</td>
```

```
            <td>${record.registerNumber}</td>
            <td>${record.status}</td>
            <td>${percentage.toFixed(2)}%</td>
        `;

        historyBody.appendChild(row);
    });
}

/* ================================
   Search Student
   ================================ */

const searchForm =
    document.getElementById("searchForm");

if (searchForm) {
    searchForm.addEventListener("submit", function (event) {

        event.preventDefault();

        const searchValue =
            document
```

```
        .getElementById("searchStudent")
        .value
        .trim()
        .toLowerCase();


const result =
    document.getElementById("searchResult");

if (!searchValue) {
    result.innerHTML =
        "<p>Please enter a student name or register number.</p>";
    return;
}


const foundStudents = students.filter(student =>
    student.name.toLowerCase().includes(searchValue) ||
    student.registerNumber.toLowerCase().includes(searchValue)
);


if (foundStudents.length === 0) {


    result.innerHTML =
        "<p>No student found.</p>";
```

```
    return;
}

result.innerHTML = "";

foundStudents.forEach(student => {

const percentage =
    student.totalClasses > 0
        ? (student.presentClasses /
            student.totalClasses) * 100
        : 0;


const box =
    document.createElement("div");

box.innerHTML = `
    <h3>${student.name}</h3>


    <p>
        Register Number:
        ${student.registerNumber}
    </p>
```

```
<p>
  Department:
  ${student.department}
</p>

<p>
  Class:
  ${student.className}
</p>

<p>
  Present:
  ${student.presentClasses}
</p>

<p>
  Absent:
  ${student.absentClasses}
</p>

<p>
  Attendance:
  ${percentage.toFixed(2)}%
</p>
```

```
            <hr>
        `;


        result.appendChild(box);
    });
  });
}

/* ================================
   Dashboard
   ================================ */

function updateDashboard() {

  const totalStudents =
    document.getElementById("totalStudents");

const presentStudents =
    document.getElementById("presentStudents");

const absentStudents =
    document.getElementById("absentStudents");
```

```
const averageAttendance =
    document.getElementById("averageAttendance");


if (totalStudents) {
    totalStudents.textContent =
        students.length;
}

let totalPresent = 0;
let totalAbsent = 0;
let totalClasses = 0;


students.forEach(student => {

    totalPresent += student.presentClasses;
    totalAbsent += student.absentClasses;
    totalClasses += student.totalClasses;
});


if (presentStudents) {
    presentStudents.textContent =
        totalPresent;
}
```

```
if (absentStudents) {
    absentStudents.textContent =
        totalAbsent;
}


let average = 0;

if (totalClasses > 0) {
    average =
        (totalPresent / totalClasses) * 100;
}


if (averageAttendance) {
    averageAttendance.textContent =
        average.toFixed(2) + "%";
}
}


/* ==================================
   Generate Report
   ================================== */


const generateReport =
```

```
    document.getElementById("generateReport");

if (generateReport) {

    generateReport.addEventListener("click", function () {

        if (students.length === 0) {
            alert("No student data available.");
            return;
        }

        let report = "STUDENT ATTENDANCE REPORT\n\n";

        students.forEach((student, index) => {

            const percentage =
                student.totalClasses > 0
                    ? (student.presentClasses /
                        student.totalClasses) * 100
                    : 0;


            report +=
                (index + 1) + ". " +
                student.name +
```

```
        " | Reg No: " +
        student.registerNumber +
        " | Attendance: " +
        percentage.toFixed(2) +
        "%\n";
});


alert(report);
    });
}


/* ==================================
   Set Today's Date
   ================================== */


const attendanceDate =
    document.getElementById("attendanceDate");

if (attendanceDate) {


    const today =
        new Date().toISOString().split("T")[0];


    attendanceDate.value = today;
```

```
}

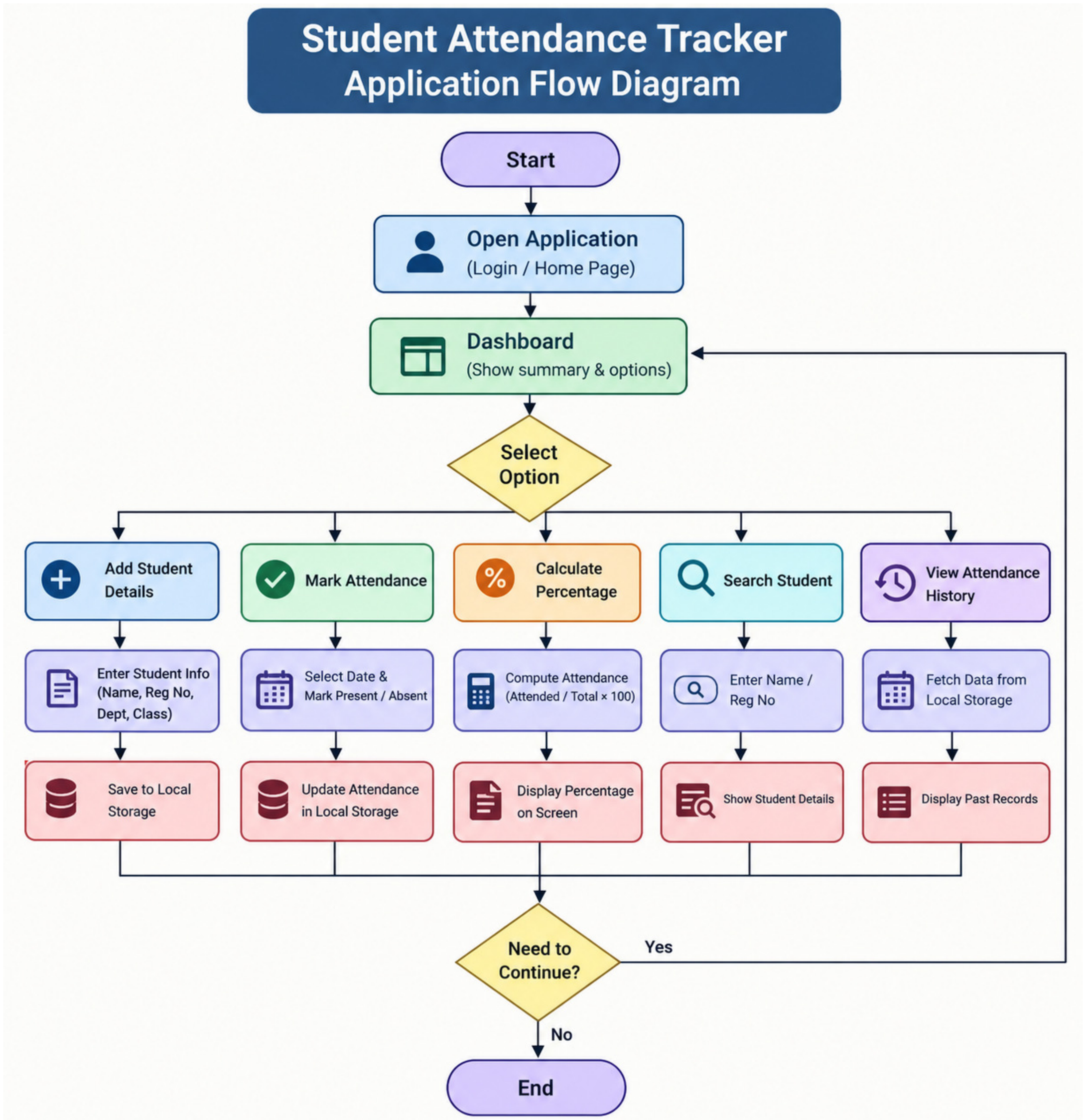
/* ==================================
   Load Data
   ================================== */

displayStudents();
displayHistory();
updateDashboard();ß
```

5. Practical Information

Feature	Implementation	Feature Description
Student Details	HTML Form	Allows faculty to enter student name, register number, department, and class details.
Attendance	JavaScript	Allows faculty to mark each student as Present or Absent.
Percentage	JavaScript	Calculate attendance percentage automatically
Search	JavaScript	Search by name or register number
Storage	Local Storage	Keep student attendance data in browser
Responsive UI	CSS	Works on desktop and mobile screens

6. Application Flow



7. Project Output / Screenshot Placeholder

figure - 1 Home/daseboard

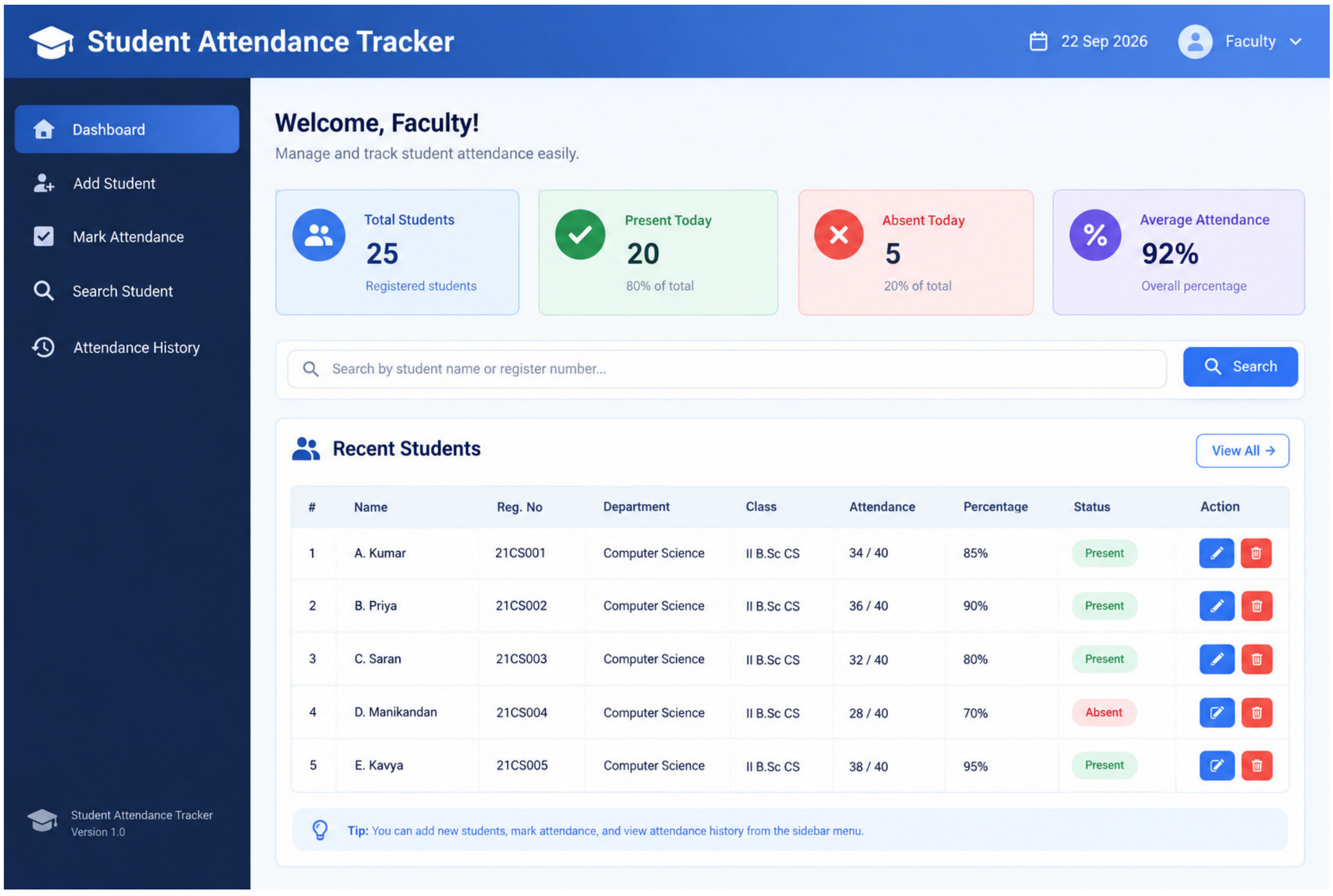


Figure 2 – Add Student Details

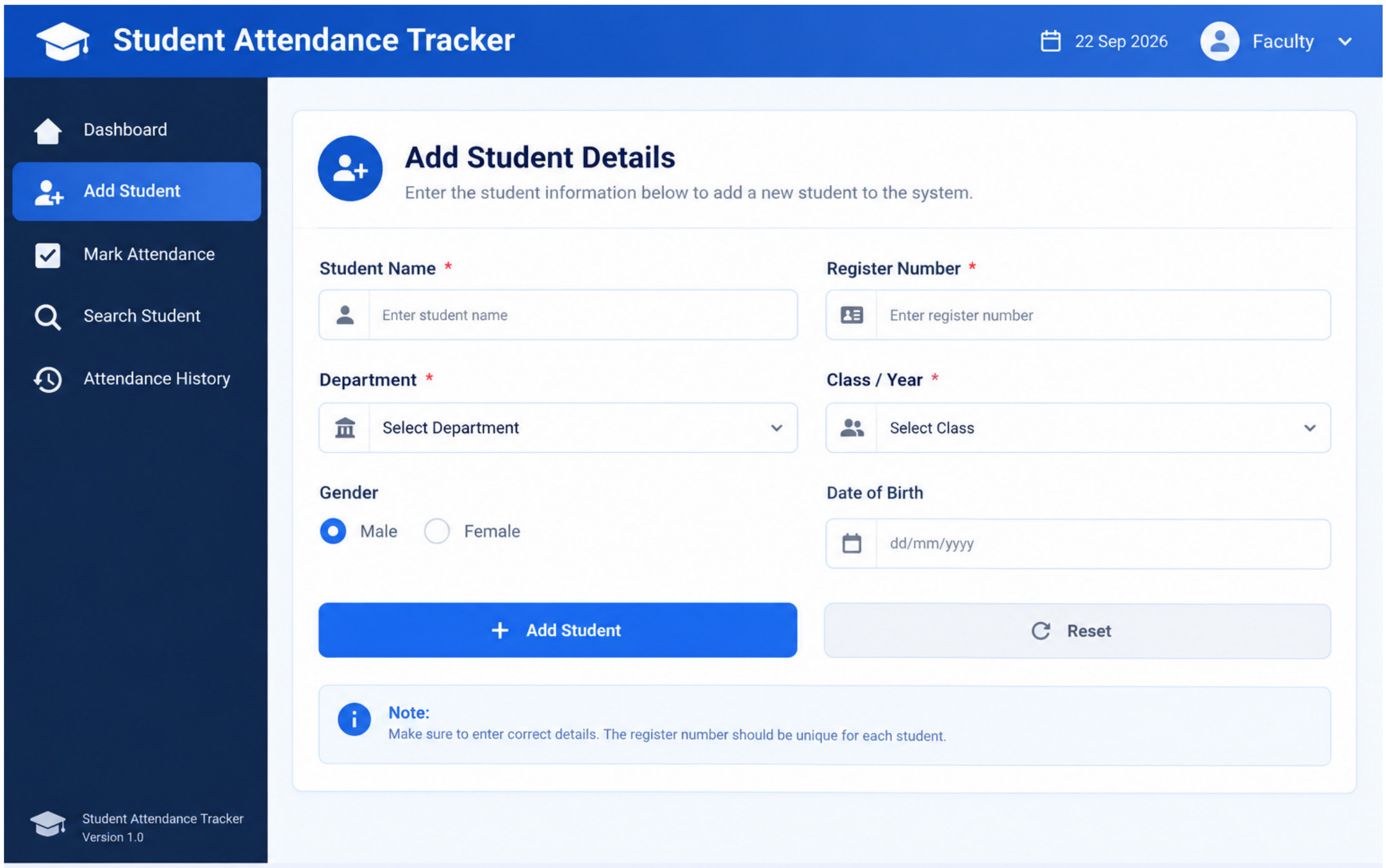


Figure 2 – Add Student Details

Figure 3 - Student Attendance List

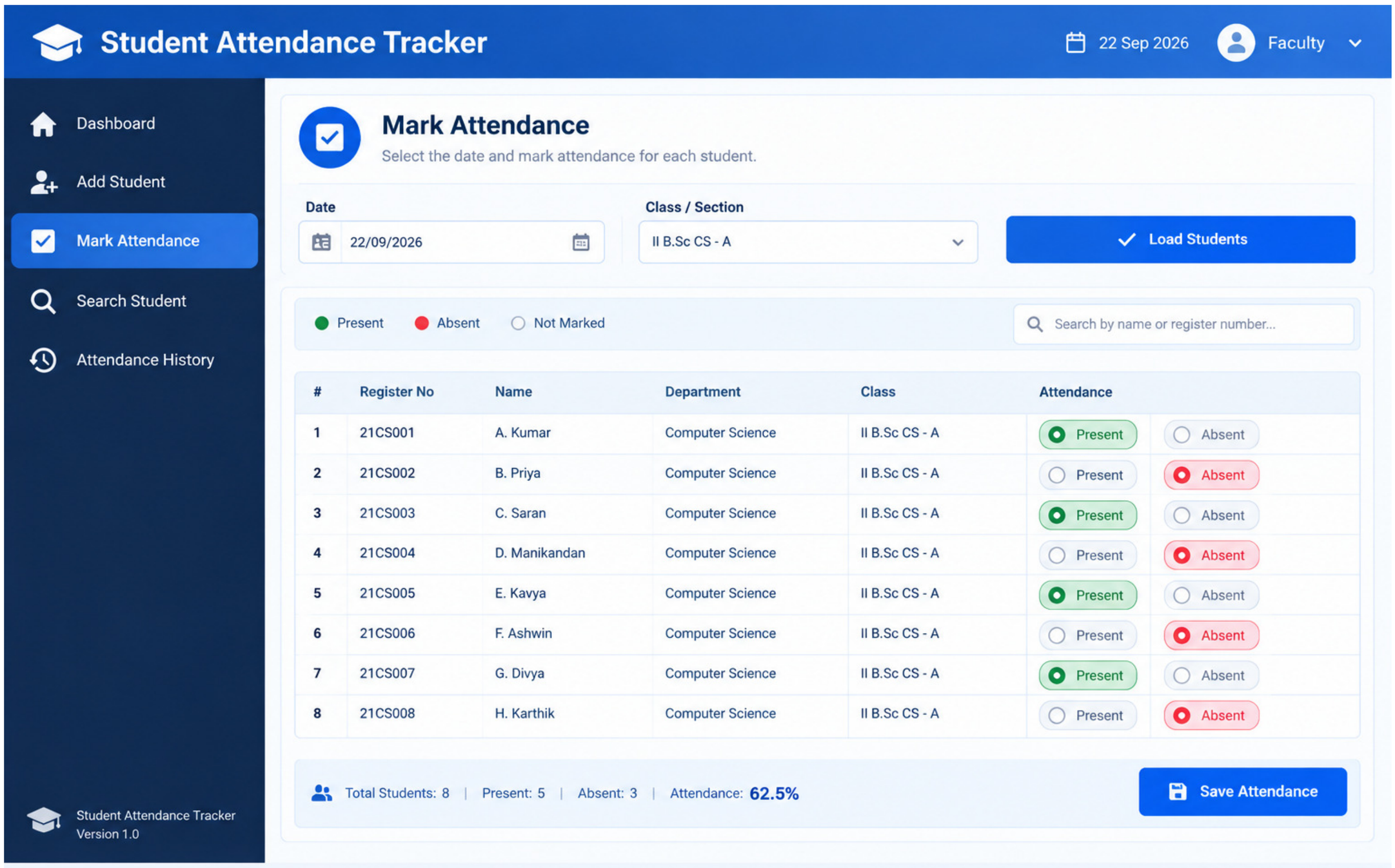


Figure 3 – Mark Attendance Screenshot

Figure 4 – Present / Absent Marking

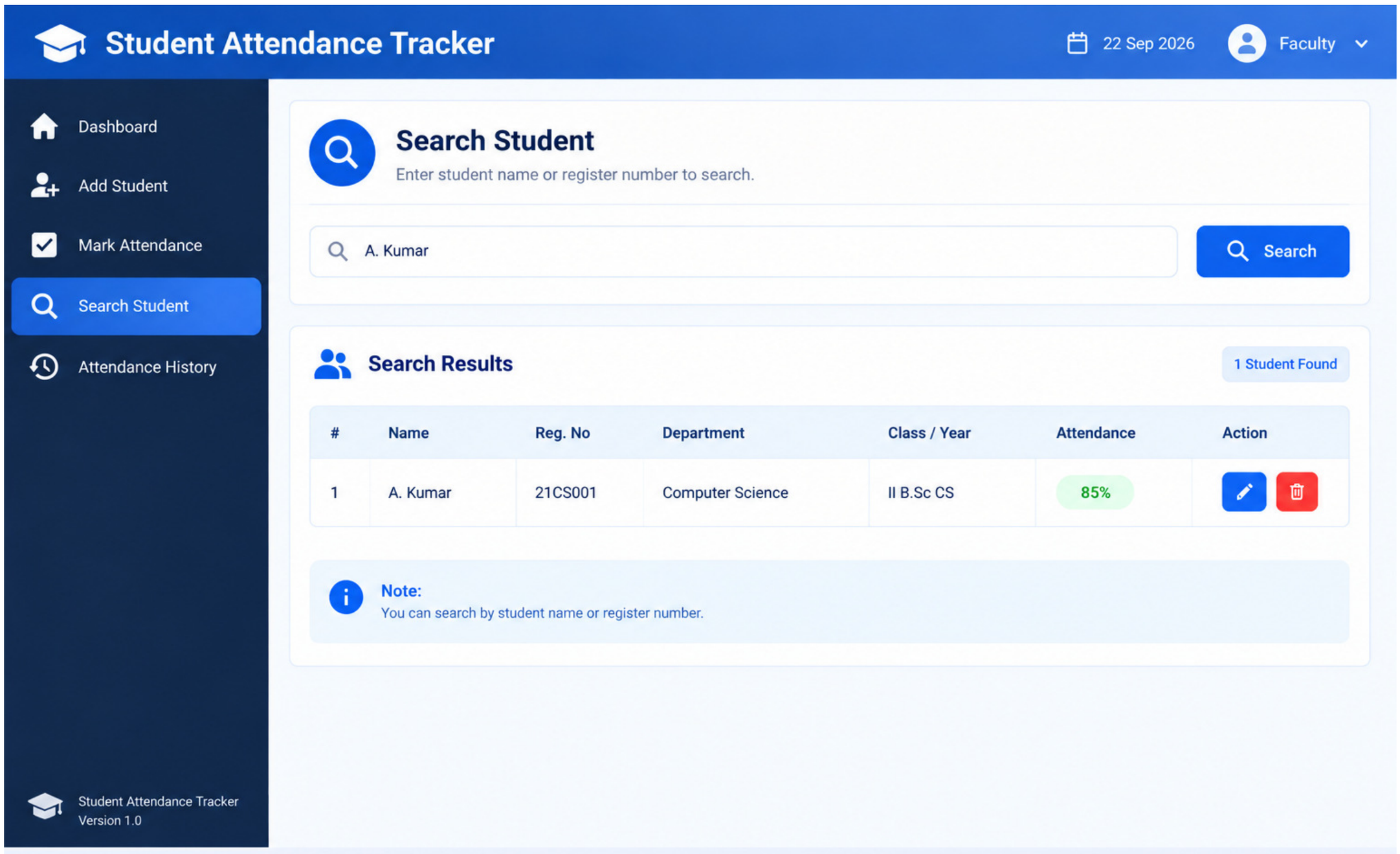


Figure 4 – Search Student Screenshot

Figure 5 – Attendance Percentage

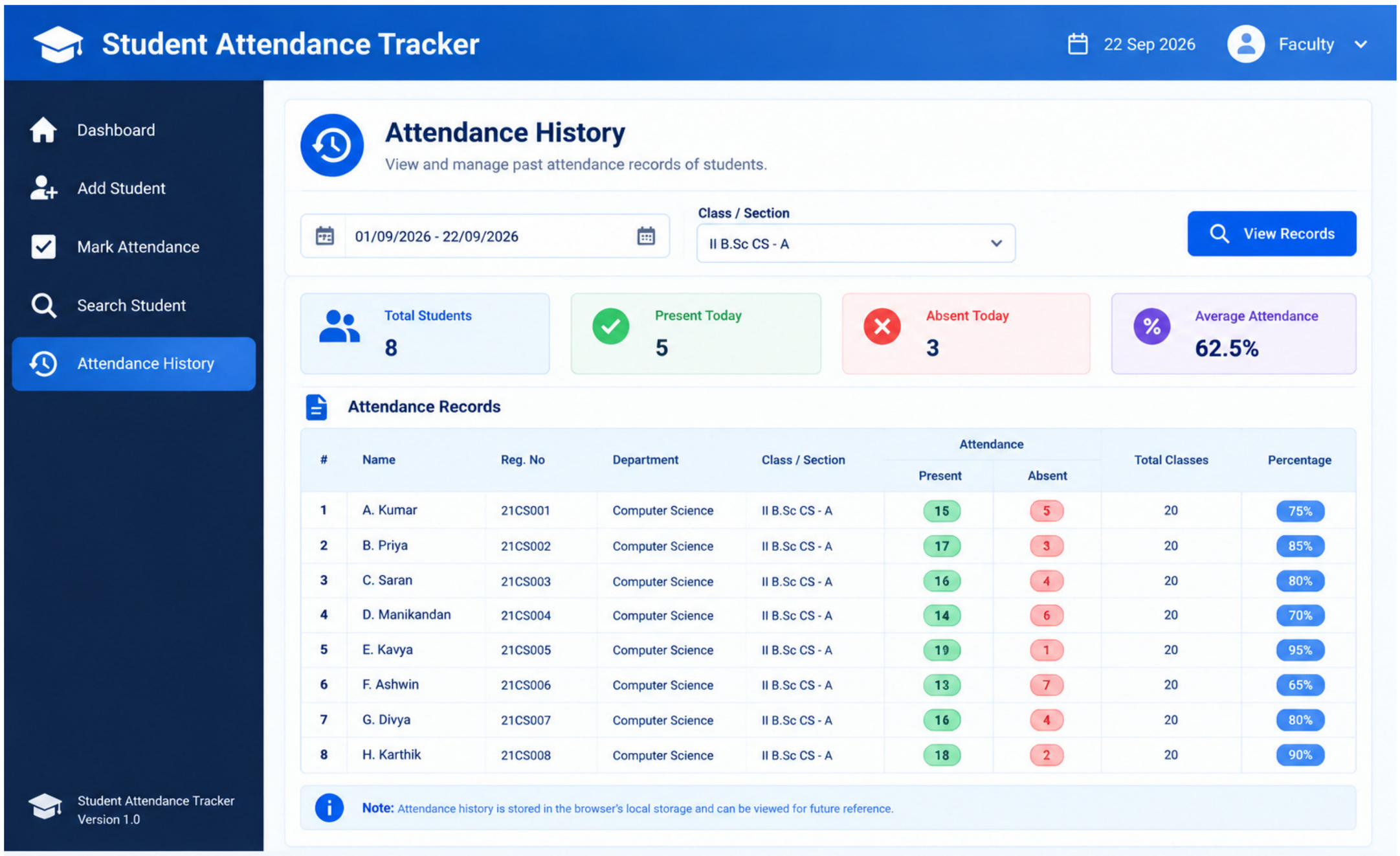


Figure 5 – Attendance History Screenshot

Figure 6 - Search / Attendance History

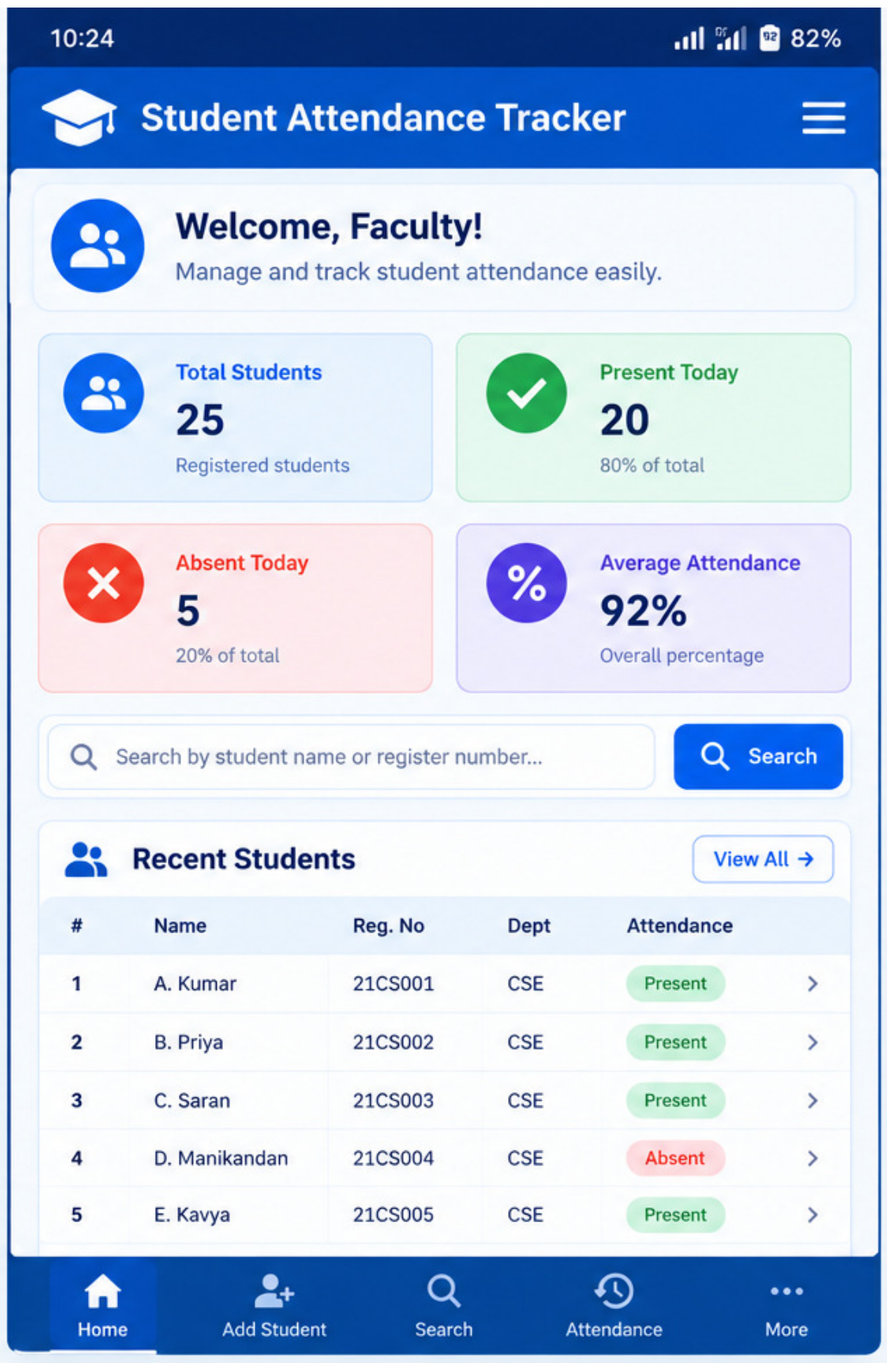


Figure 6 – Responsive Design (Mobile View) Screenshot

8. Conclusion

The Student Attendance Tracker is a simple and user-friendly web application developed to make attendance management easier and more efficient. It allows faculty to add student details, mark students as Present or Absent, calculate attendance percentages, search for students, and view attendance history. The responsive design makes the application accessible on computers, tablets, and mobile devices. Overall, the project reduces manual work, improves accuracy, and provides an organized way to monitor student attendance.

9.Future Enhancement

